

## Proving lines parallel



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- a The diagram contains parallel lines. The marked angles are supplementary and consecutive interior angles so the lines are parallel by the converse of the consecutive interior angles theorem.
- b The diagram does not contain parallel lines. The marked angles are consecutive interior angles, but not supplementary.
- c The diagram contains parallel lines. The marked angles are congruent and corresponding so the lines are parallel by the converse of the corresponding angles postulate.
- d The diagram does not contain parallel lines. The marked angles are corresponding angles, but not congruent.
- e The diagram contains parallel lines. The marked angles are congruent and alternate interior angles so the lines are parallel by the converse of the alternate interior angles theorem.
- f The diagram does not contain parallel lines. The marked angles are alternate interior, but not equal.
- g The diagram contains parallel lines. The marked angles are congruent and alternate exterior angles so the lines are parallel by the converse of the alternate exterior angles theorem.

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- a No,  $\angle CBE$  and  $\angle FEH$  are corresponding angles, and they're not congruent.
- b No,  $\angle CBE$  and  $\angle BEF$  are consecutive interior angles, and they're not supplementary.

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- a No,  $\angle ABC$  and  $\angle BCD$  are alternate interior angles, and they're not congruent.
- b Yes,  $\overline{AB}$  and  $\overline{CD}$  form supplementary consecutive interior angles with  $\overline{AC}$ .

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- a No
- b No
- c Yes
- d No

5 Consecutive interior angles converse

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To prove:  $\overleftrightarrow{AC} \parallel \overleftrightarrow{BD}$ 

| Statements                                                     | Reasons                              |
|----------------------------------------------------------------|--------------------------------------|
| 1. $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ | Given                                |
| 2. $\angle ABD \cong \angle DCA$                               | Given                                |
| 3. $\angle ABD$ and $\angle BDC$ are supplementary             | Consecutive interior angles theorem  |
| 4. $\angle DCA$ and $\angle BDC$ are supplementary             | Substitution                         |
| 5. $\overleftrightarrow{AC} \parallel \overleftrightarrow{BD}$ | Consecutive interior angles converse |

## Additional questions

7 The statement should state that  $\angle 1$  and  $\angle 2$  are supplementary instead of congruent.

8  $\angle 2$  and  $\angle 3$  are not corresponding angles.

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a  $x = \frac{7}{2}$

b The angle that forms a linear pair with the angle with measure of  $136^\circ$  would have a measure of  $44^\circ$ . If  $m\angle 1 = 44^\circ$ ,  $c \parallel d$  by the converse of the alternate interior angles theorem.

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a  $x = 25$

b  $x = 3$

c  $x = 19$

d  $x = 20$

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a Yes we can conclude  $a \parallel b$ . It is given that  $\angle 1$  and  $\angle 14$  are supplementary.  $\angle 22$  and  $\angle 16$  are supplementary by the linear pair postulate. Since  $\angle 22 \cong \angle 1$  and using substitution,  $\angle 1$  and  $\angle 16$  are supplementary. Therefore, by the congruent supplements theorem,  $\angle 14 \cong \angle 16$ . By the converse of the corresponding angles postulate,  $a \parallel b$ .

b No we can not conclude  $b \parallel c$ . We could conclude  $a \parallel b$  by the converse of the consecutive exterior angles theorem since  $\angle 10$  and  $\angle 7$  are supplementary. However, more information is needed to conclude  $b \parallel c$ .

- c Yes we can conclude  $a \parallel b$ .  $\angle 19 \cong \angle 2$  by alternate exterior angles theorem. Using substitution,  $\angle 5$  and  $\angle 2$  are supplementary. By the corresponding angles postulate  $\angle 5 \cong \angle 3$ . Using substitution,  $\angle 2$  and  $\angle 3$  are supplementary. Therefore, by the converse of the consecutive interior angles theorem,  $a \parallel b$ .
- d Yes we can conclude that  $a \parallel c$ .  $\angle 11$  and  $\angle 12$  are supplementary using the linear pair postulate. Using the congruent supplements theorem,  $\angle 1$  and  $\angle 12$  are supplementary. So by the converse of the alternate exterior angles theorem,  $a \parallel c$ .
- e Yes we can conclude  $b \parallel c$ .  $\angle 20 \cong \angle 22$  by the corresponding angles postulate.  $\angle 17$  and  $\angle 18$  are supplementary by the linear pair postulate. This means that  $\angle 17 \cong \angle 22$ . By the converse of the alternate interior angles theorem,  $b \parallel c$ .
- f No we can not conclude  $a \parallel b$ . We can conclude that  $b \parallel c$  by the converse of the alternate exterior angles theorem since  $\angle 3 \cong \angle 24$ .

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a  $\angle BMA, \angle NMS$

b

| To prove: $\overleftrightarrow{BS} \parallel \overleftrightarrow{CR}$ |                                    |
|-----------------------------------------------------------------------|------------------------------------|
| Statements                                                            | Reasons                            |
| 1. $\angle BMN$ and $\angle QNR$ are supplementary                    | Given                              |
| 2. $\angle BMN$ and $\angle BMA$ are supplementary                    | Linear pair theorem                |
| 3. $\angle BMA \cong \angle QNR$                                      | Congruent supplement theorem       |
| 4. $\overleftrightarrow{BS} \parallel \overleftrightarrow{CR}$        | Alternate exterior angles converse |

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| To prove: $\overline{AD} \parallel \overline{BC}$ |                                                     |
|---------------------------------------------------|-----------------------------------------------------|
| Statements                                        | Reasons                                             |
| 1. $\angle A$ and $\angle D$ are supplementary    | Given                                               |
| 2. $\angle B \cong \angle D$                      | Given                                               |
| 3. $\angle A$ and $\angle B$ are supplementary    | Substitution                                        |
| 4. $\overline{AD} \parallel \overline{BC}$        | Converse of the consecutive interior angles theorem |

to  $\angle 3$ . By the congruent supplements theorem,  $\angle 2 \cong \angle 3$ . Therefore, by the converse of the corresponding angles postulate,  $m \parallel n$ .



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To prove:   $\cong \angle XQP$ 

| Statements                                                     | Reasons                            |
|----------------------------------------------------------------|------------------------------------|
| 1. $\overleftrightarrow{AX} \parallel \overleftrightarrow{BY}$ | Given                              |
| 2. $\overleftrightarrow{BY} \parallel \overleftrightarrow{CZ}$ | Given                              |
| 3. $\angle XQP \cong \angle YRQ$                               | Corresponding angles theorem       |
| 4. $\angle YRQ \cong \angle ZSR$                               | Corresponding angles theorem       |
| 5. $\angle XQP \cong \angle ZSR$                               | Transitive property of congruence  |
| 6. $\angle ZSR \cong \angle CST$                               | Vertical angles congruence theorem |
| 7. $\angle XQP \cong \angle CST$                               | Transitive property of congruence  |

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To prove:  $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$ 

| Statements                                                     | Reasons                                           |
|----------------------------------------------------------------|---------------------------------------------------|
| 1. $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$ | Given                                             |
| 2. $\angle NBA$ and $\angle ADC$ are supplementary             | Given                                             |
| 3. $\angle NBA$ and $\angle DCR$ are supplementary             | Consecutive exterior angles theorem               |
| 4. $\angle ADC \cong \angle DCR$                               | Congruent supplements theorem                     |
| 5. $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$ | Converse of the alternate interior angles theorem |

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To prove:  $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$ 

| Statements                                                     | Reasons                                           |
|----------------------------------------------------------------|---------------------------------------------------|
| 1. $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$ | Given                                             |
| 2. $\angle MAB \cong \angle BCQ$                               | Given                                             |
| 3. $\angle BCQ \cong \angle ABC$                               | Alternate interior angles theorem                 |
| 4. $\angle MAB \cong \angle ACB$                               | Transitive property of congruence                 |
| 5. $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$ | Converse of the alternate interior angles theorem |